

Thermal Impedance Estimate of the Clamp Heatsink 778746

PADA profile “G”, 225×91 mm section, $L = 250$ mm, forced air cooling

D. Bagnara

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Abstract

This note estimates the heatsink-to-ambient thermal resistance and the transient thermal impedance of the clamp heatsink of drawing 778746-01-01 (*Dissipatore clamp L250 225x91*), used with a press-pack thyristor having a 100 mm contact face. The heatsink is a modular aluminium extrusion (PADA profile “G”: 18 central modules plus two end modules) with two 14 mm plates joined by 36 internal fins; the fins form 37 closed channels of about 4.15×63 mm, 250 mm long, through which cooling air must be forced. Two independent models are used: an analytical chain (channel convection correlation, fin analysis, spreading resistance) and a three-dimensional finite-difference model that also provides the step response. For a single device on one face the estimated resistance is $R_{th} \approx 0.080$ K/W (centre of the contact face) at $140 \text{ m}^3/\text{h}$, decreasing to 0.068 K/W at $280 \text{ m}^3/\text{h}$; a design value with margin is proposed in Section 7. A four-cell Foster network is given for transient calculations.

1 Scope and input data

1.1 Data taken from the drawing

Table 1 lists the data read from drawing 778746-01-01 (rev. 1, 28/03/2019). Only the overall dimensions, the material, the module composition and the hole pattern are dimensioned explicitly. The internal fin geometry is drawn schematically, so fin thickness, channel width and plate thickness were scaled from the section view (225×91 mm frame, 800 dpi raster); an uncertainty of about ± 0.2 mm on these values must be expected and its effect is discussed in Section 5.

Table 1: Heatsink data. Values marked * are scaled from the section view.

Symbol	Quantity	Value
	Part / drawing no.	GR19 5000A / 778746
	Material	Al alloy for extrusion UNI 9006/1 (6060/6063)
	Construction (PADA profile “G”)	G8716 $\times$ 1, G8717 $\times$ 1, G8715 $\times$ 18
W	Width across the fins	225 mm
H_s	Overall height (plate to plate)	91 mm
L	Length along the channels (extrusion axis)	250 mm
D_s	SCR contact area (one face, centred, R_a 1.2)	$\varnothing$ 100 mm
	Clamp tie-rod holes (centre line)	$2 \times \varnothing$ 18 mm through, 180 mm apart
t_b	Plate thickness (surface to fin root), each side*	14 mm
N_f	Internal fins (webs)*	36
t_f	Fin thickness*	1.6 mm
b	Channel width*	4.15 mm
p	Fin pitch* ($= t_f + b$)	5.75 mm
N_{ch}	Channels* (4.15×63 mm, rounded ends)	37
t_w	Side wall thickness*	6.9 mm

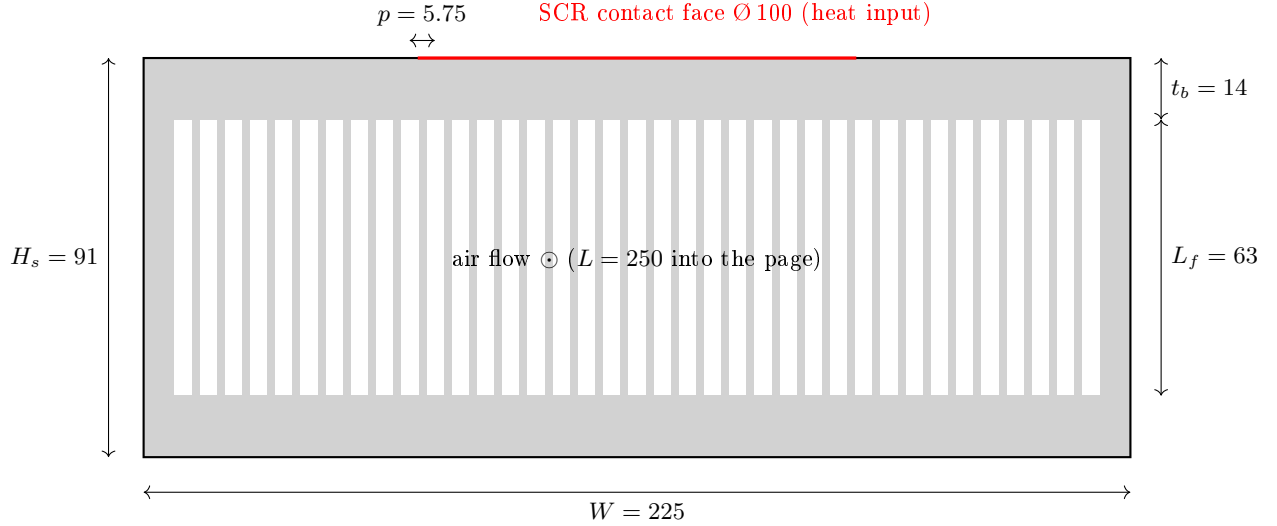


Figure 1: Schematic cross-section used for the thermal model (dimensions in mm). The two 14 mm plates are connected by 36 fins of 1.6 mm; the 37 channels are closed on all four sides. The drawing shows the fin surfaces as corrugated; a smooth surface is assumed here (conservative).

1.2 Derived geometric quantities

$$\begin{aligned}
 L_f &= H_s - 2t_b = 63 \text{ mm} & A_s &= \frac{\pi}{4} D_s^2 = 7854 \text{ mm}^2 \\
 A_p &= WL = 56\,250 \text{ mm}^2 & A_s/A_p &= 0.14 \\
 A_{ff} &= N_{ch} b L_f = 96.7 \text{ cm}^2 & \sigma &= A_{ff}/(WH_s) = 0.47 \\
 D_h &= 2bL_f/(b + L_f) = 7.8 \text{ mm} & A_{fin} &= 2N_f L_f L = 1.13 \text{ m}^2
 \end{aligned}$$

The exposed channel top and bottom walls add $N_{ch}bL = 0.038 \text{ m}^2$ each and the inner faces of the side walls 0.032 m^2 . The metal volume is about 2.7 L, i.e. a mass of about 7.3 kg and a total heat capacity of about 6.6 kJ/K.

1.3 Assumptions

1. Forced air enters one 225×91 mm face and leaves the opposite one; all channels receive the same flow (no bypass, no maldistribution). Results are given as a function of the volumetric flow Q actually passing through the heatsink and of the corresponding channel velocity $V_{ch} = Q/A_{ff}$.
2. One thyristor on one face, uniform heat flux over the $\varnothing 100$ mm contact area. The device-to-heatsink contact resistance $R_{th}(c-h)$ is *not* included; it is taken from the device datasheet.
3. Aluminium: $k = 200 \text{ W/(m K)}$, $\rho = 2700 \text{ kg/m}^3$, $c_p = 900 \text{ J/(kg K)}$. Air at 40°C : $\rho_a = 1.127 \text{ kg/m}^3$, $c_{p,a} = 1007 \text{ J/(kg K)}$, $k_a = 0.0271 \text{ W/(m K)}$, $\nu_a = 1.70 \times 10^{-5} \text{ m}^2/\text{s}$, $\text{Pr} = 0.71$.
4. Baseline: the plates are treated as monolithic. The module joints (one every two fins) are examined as a sensitivity in Section 5.
5. Outer faces adiabatic (the second plate is not cooled from outside), radiation neglected, smooth fin surfaces, tie-rod holes neglected (they remove less than 1 % of the fin surface).
6. The convection coefficient is referred to the *inlet* air temperature, so that all resistances are heatsink-to-inlet-air values; the air temperature rise along the channels is embedded in the correlation of Section 2.1.

2 Analytical steady-state model

The thermal path is represented by the series of a spreading/conduction resistance in the loaded plate and a convective resistance of the fin array (Fig. 2):

$$R_{\text{th(h-a)}} = R_{sp} + \frac{1}{G}, \quad G = N_f G_f + 2G_w + h N_{ch} b L. \quad (1)$$

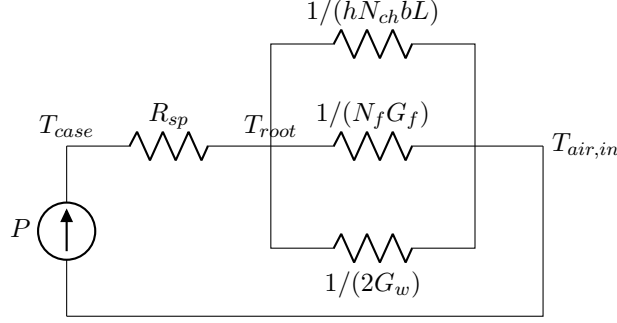


Figure 2: Steady-state network of Eq. (1): plate spreading in series with the parallel of fins, side walls and directly exposed channel top walls.

2.1 Channel convection coefficient

The channels are narrow ($b/L_f = 0.066$) and the flow is laminar over the whole practical range ($\text{Re}_{D_h} < 2500$ up to 10 m/s). With $L/D_h = 32$ the thermal boundary layers are still developing over a large part of the length, so neither the fully developed nor the flat-plate limit applies alone. The composite model of Teertstra, Yovanovich and Culham [1] for parallel-plate channels is used:

$$\text{Nu}_b = \left[\left(\frac{\text{Re}_b^* \text{Pr}}{2} \right)^{-3} + \left(0.664 \sqrt{\text{Re}_b^*} \text{Pr}^{1/3} \sqrt{1 + \frac{3.65}{\sqrt{\text{Re}_b^*}}} \right)^{-3} \right]^{-1/3}, \quad \text{Re}_b^* = \frac{V_{ch} b}{\nu_a} \cdot \frac{b}{L}, \quad (2)$$

and $h = \text{Nu}_b k_a / b$. The first term is the energy-balance limit (outlet air at wall temperature), the second the developing-flow limit; h is therefore defined on the wall-to-inlet temperature difference. At $V_{ch} = 4$ m/s ($Q \approx 140$ m³/h): $\text{Re}_b = 976$, $\text{Re}_b^* = 16$, $\text{Nu}_b = 3.1$, $h = 20$ W/(m² K). The heatsink then works at about 55 % of its energy-balance limit, i.e. the outlet air is already at roughly half of the wall-to-inlet temperature difference: increasing the flow is more effective than increasing the surface.

2.2 Fin conductance

Each fin (t_f , length L_f , depth L) is a straight fin whose far end is attached to the second plate. The second plate is thick and nearly isothermal; it collects the fin-tip heat and releases it through the channel bottom walls. It is therefore modelled as a convective tip whose coefficient is scaled by the collector-to-cross-section area ratio, $h_t = h b / t_f$, giving

$$G_f = \sqrt{2hkt_f} L \frac{\sinh(mL_f) + \zeta \cosh(mL_f)}{\cosh(mL_f) + \zeta \sinh(mL_f)}, \quad m = \sqrt{\frac{2h}{kt_f}}, \quad \zeta = \frac{h_t}{mk}. \quad (3)$$

The same expression, with one convecting face and $h_t = h (b/2) / t_w$, gives the side-wall conductance G_w . At $h = 20$ W/(m² K): $mL_f = 0.71$, adiabatic-tip efficiency 0.86, effective efficiency including the second plate $\eta_f = G_f / (2hL_f L) = 0.88$. The fins are efficient; a thicker fin would gain little.

2.3 Spreading resistance in the loaded plate

The $\varnothing 100$ mm source covers only 14 % of the 250×225 mm plate, and the plate (14 mm) is thin compared with the spreading distance, so the constriction resistance is significant. The closed-form model of Lee, Song, Au and Moran [2] for a circular source of radius a on a circular plate of equivalent radius $r_p = \sqrt{A_p/\pi} = 133.8$ mm, thickness t_b , with a uniform effective coefficient $h_{eff} = G/A_p$ on the finned side, is used:

$$\begin{aligned} \varepsilon &= \frac{a}{r_p}, \quad \tau = \frac{t_b}{r_p}, \quad \text{Bi} = \frac{h_{eff} r_p}{k}, \quad \lambda = \pi + \frac{1}{\varepsilon\sqrt{\pi}}, \quad \Phi = \frac{\tanh(\lambda\tau) + \lambda/\text{Bi}}{1 + (\lambda/\text{Bi}) \tanh(\lambda\tau)}, \\ \psi_{max} &= \frac{\varepsilon\tau}{\sqrt{\pi}} + \frac{(1-\varepsilon)\Phi}{\sqrt{\pi}}, \quad \psi_{avg} = \frac{\varepsilon\tau}{\sqrt{\pi}} + \frac{1}{2}(1-\varepsilon)^{3/2}\Phi, \quad R_{sp} = \frac{\psi}{k\sqrt{A_s}}. \end{aligned} \quad (4)$$

ψ_{max} refers to the centre of the contact face, ψ_{avg} to its area-averaged temperature; both include the one-dimensional conduction through the plate. At 4 m/s: $\text{Bi} = 0.26$, $\Phi = 2.0$, $R_{sp,max} = 0.042$ K/W, $R_{sp,avg} = 0.030$ K/W, against a convective resistance $1/G = 0.046$ K/W: the plate constriction is of the same order as the convective term and is almost independent of the flow rate. This is the main limitation of the heatsink for a $\varnothing 100$ mm device.

2.4 Pressure drop

For fan selection the pressure drop across the heatsink is estimated with the laminar developing-flow apparent friction factor of Shah and London [3] for a rectangular duct of aspect ratio b/L_f , plus contraction and expansion losses at the faces:

$$\Delta p = \frac{1}{2}\rho_a V_{ch}^2 \left[4f_{app} \frac{L}{D_h} + K_c + K_e \right], \quad K_c = 0.42(1 - \sigma^2), \quad K_e = (1 - \sigma^2)^2. \quad (5)$$

Duct and plenum losses are not included.

3 Finite-difference model

The analytical chain treats the plate as isotropic and the fin array as a uniform film. In reality the fins are continuous plates in the flow direction and conduct heat along x , the second plate spreads heat too, and the fin-root temperature is far from uniform. To check the analytical result and to obtain the step response, a three-dimensional finite-difference model was built (script in Appendix B):

- loaded plate: $50 \times 38 \times 7$ cells ($\Delta x = 5$ mm, one cell per fin pitch in y , $\Delta z = 2$ mm);
- fins and side walls: 50×10 cells each, conduction along x and along the fin, convection on both faces (one face for the side walls);
- second plate: $50 \times 38 \times 2$ cells, fed by the fin tips, convecting through the channel bottom walls;
- source: uniform flux on the top layer over the cells inside the $\varnothing 100$ mm circle (area fractions by sub-sampling);
- uniform h from Eq. (2), referred to inlet air; 36 100 unknowns.

The steady state is solved directly; the step response is integrated with the implicit Euler scheme (90 steps per decade from 1 ms to 10^5 s). The reported case temperature is the surface value (node value plus half-cell correction), either averaged over the contact area or taken at its centre.

Figure 3 shows the temperature distribution at 140 m³/h. The fin-root temperature under the centre of the device is about 1.8 times the value at the plate ends; the fins directly under the footprint (14 % of the area) convect only 17.5 % of the heat, i.e. the plate does spread the heat, but at the cost of a large lateral temperature drop. The second plate sits at roughly half of the loaded-plate centre temperature rise: the fins work as a bridge between the two plates and the second plate contributes to the exchange.

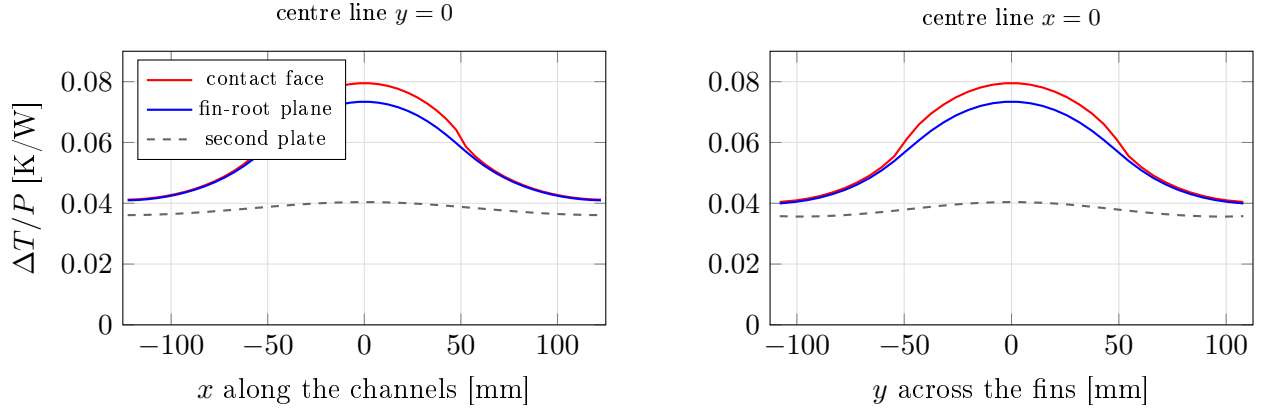


Figure 3: Steady-state temperature rise above inlet air per watt (finite-difference model, $Q = 139 \text{ m}^3/\text{h}$, $V_{ch} = 4 \text{ m/s}$). The source extends over $|x|, |y| < 50 \text{ mm}$.

4 Steady-state results

Table 2 and Fig. 4 give the results of both models as a function of the flow through the heatsink. The finite-difference values are 6 % to 10 % lower than the analytical ones because of the additional spreading through the fins and the second plate; the agreement supports both estimates. The finite-difference value referred to the centre of the contact face, R_{th}^{max} , is the recommended basis since the case temperature of a press-pack device is normally checked at the centre of the pole face; R_{th}^{avg} is the area-averaged value.

Table 2: Heatsink-to-inlet-air resistance versus flow. Q : flow through the heatsink; V_{ch} : channel velocity; η_f : effective fin efficiency; G : total conductance of the fin array; R_{sp} : plate spreading (Eq. 4); “an.”: Eq. (1); “FD”: finite-difference model; Δp : Eq. (5). Resistances in K/W.

Q m^3/h	V_{ch} m/s	Re_b	Nu_b	h $\text{W}/(\text{m}^2 \text{ K})$	η_f	G W/K	$1/G$	R_{sp}^{avg}	R_{sp}^{max}	R_{th}^{avg} an.	R_{th}^{max} an.	R_{th}^{avg} FD	R_{th}^{max} FD	Δp Pa
35	1.0	244	1.29	8.5	0.962	9.8	0.1018	0.0309	0.0435	0.1327	0.1454	0.1270	0.1358	4
52	1.5	366	1.77	11.5	0.939	13.1	0.0764	0.0305	0.0430	0.1069	0.1193	0.1015	0.1102	7
70	2.0	488	2.14	14.0	0.922	15.6	0.0642	0.0302	0.0425	0.0944	0.1067	0.0892	0.0979	11
87	2.5	610	2.44	15.9	0.909	17.5	0.0570	0.0300	0.0422	0.0870	0.0993	0.0820	0.0906	14
104	3.0	732	2.69	17.6	0.898	19.1	0.0522	0.0298	0.0420	0.0820	0.0942	0.0771	0.0858	18
139	4.0	976	3.11	20.3	0.881	21.7	0.0461	0.0295	0.0415	0.0756	0.0876	0.0708	0.0795	28
174	5.0	1221	3.45	22.5	0.867	23.7	0.0421	0.0293	0.0412	0.0714	0.0834	0.0668	0.0754	39
209	6.0	1465	3.75	24.5	0.856	25.5	0.0393	0.0291	0.0410	0.0684	0.0803	0.0639	0.0725	53
244	7.0	1709	4.01	26.2	0.846	26.9	0.0371	0.0289	0.0407	0.0661	0.0779	0.0617	0.0702	68
279	8.0	1953	4.25	27.7	0.837	28.3	0.0354	0.0288	0.0405	0.0642	0.0759	0.0599	0.0684	84
348	10.0	2441	4.68	30.5	0.822	30.6	0.0327	0.0286	0.0402	0.0612	0.0729	0.0571	0.0656	122

For quick use, the finite-difference results are fitted within 1.5 % over $35 \text{ m}^3/\text{h}$ to $350 \text{ m}^3/\text{h}$ by

$$R_{th}^{max}(\text{h-a}) \approx 0.057 + 2.14 Q^{-0.93}, \quad R_{th}^{avg}(\text{h-a}) \approx 0.049 + 2.17 Q^{-0.94} \quad [\text{K/W}, Q \text{ in } \text{m}^3/\text{h}]. \quad (6)$$

The constant term is essentially the plate spreading resistance: even with unlimited flow the resistance would not fall below about 0.05 K/W , which sets the useful range of this heatsink with a $\varnothing 100 \text{ mm}$ device.

5 Sensitivity and uncertainty

Module joints. The plates are built from 20 interlocked modules; in the finite-difference model a contact conductance h_j was inserted at every module boundary (every two fins) across the full plate thickness, Table 3. Pressed aluminium-to-aluminium joints typically lie between $10 \text{ kW}/(\text{m}^2 \text{ K})$ and $50 \text{ kW}/(\text{m}^2 \text{ K})$; the hooked interface has about twice the area of a plane cut, which partly

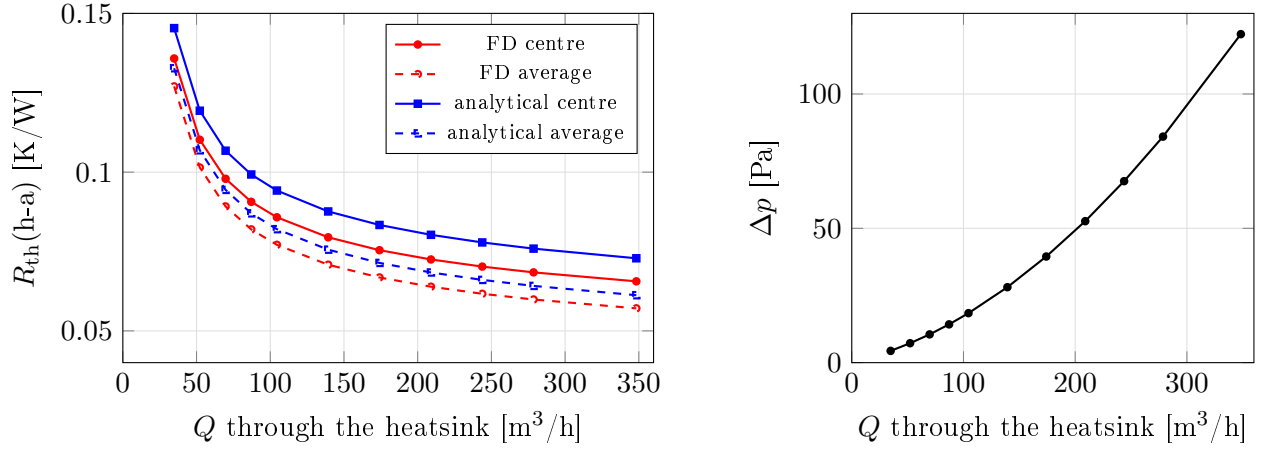


Figure 4: Left: heatsink-to-inlet-air resistance versus flow. Right: estimated pressure drop across the heatsink (channels only).

compensates. An increase of 5 % to 15 % of the resistance with respect to the monolithic case should be expected unless the modules are bonded.

Table 3: Effect of the module joint conductance at $Q = 139 m^3/h$ (FD model).

h_j [kW/(m ² K)]	R_{th}^{avg}	R_{th}^{max}	ΔR_{th}^{avg}	ΔR_{th}^{max}
∞ (monolithic)	0.0708	0.0795	+0 %	+0 %
50	0.0734	0.0830	+4 %	+4 %
20	0.0765	0.0873	+8 %	+10 %
10	0.0806	0.0929	+14 %	+17 %
5	0.0866	0.1010	+22 %	+27 %

Other parameters (at $139 m^3/h$).

- Aluminium conductivity $170 W/(m K)$ instead of 200 (e.g. 6082 or poorly tempered 6060): +8 % on R_{th}^{max} .
- Fin thickness $1.4 mm$ instead of 1.6 : η_f drops from 0.88 to 0.87 , effect below 2 %. Channel width $\pm 0.2 mm$: changes V_{ch} at constant Q by ± 5 %, effect on R_{th} about ± 2 %.
- Natural convection ($5 W/(m^2 K)$) on the free face of the second plate and on the side walls: $R_{th}^{avg} = 0.0703 K/W$, $R_{th}^{max} = 0.0789 K/W$, i.e. less than 1 % gain; the outer faces do not matter.
- Corrugated fin surfaces, as suggested by the drawing, would increase h by typically 20 % to 40 % in this laminar regime; since half of the resistance is in the plate, the gain on R_{th} would be limited to 8 % to 15 %. It is not credited here.
- Flow maldistribution or bypass around the heatsink reduce the effective Q ; Eq. (6) should be entered with the flow actually passing through the channels.

Overall. Correlation-based estimates of this kind are usually within ± 15 % of measurements. Combining this with the joint effect, a design margin of 15 % on the monolithic finite-difference value is suggested.

Natural convection. With the channels vertical (Elenbaas correlation [5], $\Delta T = 50$ K) the channel coefficient would be only about $1.2 \text{ W}/(\text{m}^2 \text{ K})$ because of the 4 mm gap; including the outer faces the resistance would be of the order of $0.4 \text{ K}/\text{W}$ to $0.6 \text{ K}/\text{W}$. The heatsink is not meant for natural convection.

6 Transient thermal impedance

Figure 5 shows the step response of the finite-difference model for three flow rates; Table 4 lists selected values. Up to about 30 s the impedance is independent of the flow (within 1.5 %) and up to 100 s within 5 %: overload calculations in this range do not depend on the fan operating point. The steady state is reached after about one hour ($\tau \approx 170$ s to 370 s for the last cell, set by the $6.6 \text{ kJ}/\text{K}$ of the aluminium).

For $t < 0.1$ s the mesh resolution (2 mm layers) is not sufficient; there the semi-infinite-body solution applies,

$$Z_{\text{th}}(t) \approx \frac{2}{kA_s} \sqrt{\frac{\alpha t}{\pi}}, \quad \alpha = \frac{k}{\rho c_p} = 8.2 \times 10^{-5} \text{ m}^2/\text{s}, \quad (7)$$

which gives $0.65 \times 10^{-3} \text{ K}/\text{W}$ at 10 ms and matches the model from 0.1 s on (dotted line in Fig. 5). In this range the heatsink contribution is negligible compared with the junction-to-case impedance of the device.

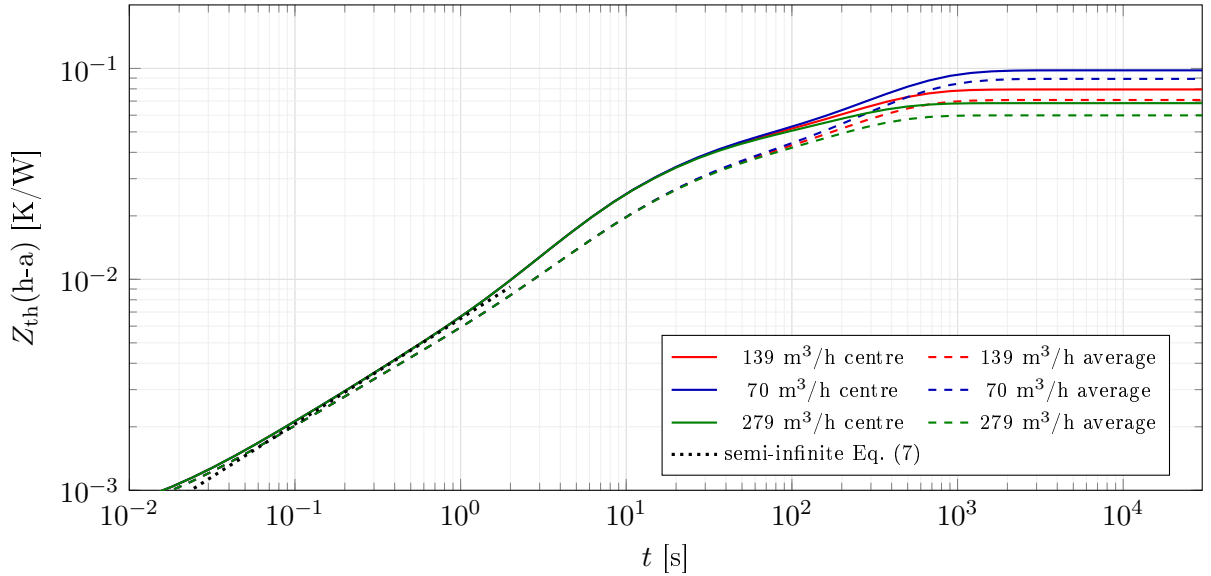


Figure 5: Transient thermal impedance heatsink-to-inlet-air (step response, finite-difference model).

6.1 Foster network

For circuit or system simulation the curves were fitted from 50 ms upwards with four Foster cells,

$$Z_{\text{th}}(t) = \sum_{i=1}^4 R_i \left(1 - e^{-t/\tau_i}\right), \quad C_i = \tau_i/R_i, \quad (8)$$

Table 5 (maximum deviation 5 % of the local value, 1.4 % of the final value). The first three cells are practically flow independent; only R_4 and τ_4 change with Q . For a flow other than the tabulated ones, keep cells 1–3 of the $139 \text{ m}^3/\text{h}$ set and take $R_4 = R_{\text{th}}(Q) - R_1 - R_2 - R_3$ with $\tau_4 \approx 6.0 \text{ kJ}/\text{K} \cdot R_4$.

Table 4: $Z_{\text{th}}(\text{h-a})$ in mK/W at selected times (FD model).

t [s]	70 m ³ /h		139 m ³ /h		279 m ³ /h	
	avg	centre	avg	centre	avg	centre
0.1	2.0	2.1	2.0	2.1	2.0	2.1
0.3	3.4	3.6	3.4	3.6	3.4	3.6
1	5.9	6.6	5.9	6.6	5.9	6.6
3	10.5	12.7	10.5	12.7	10.5	12.7
10	19.8	25.5	19.8	25.4	19.7	25.4
30	31.1	39.1	30.9	38.9	30.7	38.6
100	44.2	53.0	43.2	51.9	42.1	50.7
300	62.4	71.2	57.7	66.4	53.4	61.9
1000	84.8	93.5	69.9	78.5	59.7	68.2
3000	89.2	97.9	70.8	79.5	59.9	68.4

Table 5: Four-cell Foster parameters of $Z_{\text{th}}(\text{h-a})$. R_i in mK/W, τ_i in s, C_i in J/K.

i	contact-area average			contact-area centre		
	R_i	τ_i	C_i	R_i	τ_i	C_i
$Q = 70 \text{ m}^3/\text{h}$ ($V_{ch} = 2 \text{ m/s}$)						
1	1.65	0.04	27	1.74	0.05	27
2	3.27	0.79	240	3.09	0.86	278
3	24.68	12.89	522	33.04	11.59	351
4	59.49	371.55	6246	59.91	366.32	6114
$\sum R_i$	89.1			97.8		
$Q = 139 \text{ m}^3/\text{h}$ ($V_{ch} = 4 \text{ m/s}$)						
1	1.61	0.04	27	1.63	0.04	26
2	3.14	0.73	232	2.69	0.67	249
3	24.18	12.31	509	32.51	10.85	334
4	41.81	250.64	5995	42.51	242.93	5715
$\sum R_i$	70.7			79.3		
$Q = 279 \text{ m}^3/\text{h}$ ($V_{ch} = 8 \text{ m/s}$)						
1	1.56	0.04	26	1.47	0.04	25
2	2.97	0.66	221	2.37	0.49	208
3	23.45	11.58	494	31.46	10.00	318
4	31.80	176.97	5564	32.97	166.10	5037
$\sum R_i$	59.8			68.3		

6.2 Use with the device data

The junction temperature follows from the series chain

$$T_j(t) = T_{air,in} + P * [Z_{th}(j-c) + R_{th}(c-h) + Z_{th}(h-a)], \quad (9)$$

where $Z_{th}(j-c)$ and the contact resistance $R_{th}(c-h)$ (single-side cooled value) are taken from the thyristor datasheet and $*$ denotes convolution. For a device with a $\varnothing 100$ mm pole face $R_{th}(c-h)$ is typically 3 mK/W to 5 mK/W per side, i.e. small compared with the heatsink. If two devices are mounted, one on each face, each device sees approximately the half-heatsink (fins with adiabatic mid-plane); the per-device resistance is then about 1.4–1.6 times the values given here (analytical estimate, higher factor at low flow) rather than twice, because the spreading resistance does not double.

7 Summary

Table 6: Proposed design values for heatsink 778746, single device, contact-face centre, including 15 % margin on the finite-difference result.

Q [m ³ /h]	V_{ch} [m/s]	Δp [Pa]	R_{th}^{max} FD [K/W]	design $R_{th}(h-a)$ [K/W]
70	2	11	0.098	0.113
104	3	18	0.086	0.099
139	4	28	0.080	0.091
209	6	53	0.073	0.083
279	8	84	0.068	0.079

- The heatsink is a closed-channel forced-air design; usable performance requires a fan delivering 100 m³/h to 300 m³/h *through* the heatsink at 20 Pa to 90 Pa.
- The resistance is split roughly half and half between plate spreading (flow independent, about 0.04 K/W at the centre) and convection. Above 200 m³/h the return on additional flow is small.
- With a 60 K allowed rise above inlet air, the dissipable power is about 650 W to 750 W per heatsink in the 140 m³/h to 280 m³/h range.
- For overloads shorter than about 30 s the transient impedance does not depend on the flow; the Foster set of Table 5 can be used directly.
- The estimate should be confirmed by a measurement on a sample or by manufacturer data for the profile “G” assembly, in particular for the behaviour of the module joints.

References

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A Analytical model script

Python 3 with NumPy; reproduces Table 2 (analytical columns).

```
"""
Heatsink 778746 (SECOM, PADA profile "G", L = 250 mm) -- analytical steady-state model.
Forced convection through the internal channels, fin analysis, Lee spreading resistance,
pressure drop. Reproduces Table 3 of the accompanying note.
"""
import numpy as np

# ---- geometry [m] (from drawing 778746-01-01) ----
W, Hs, L = 0.225, 0.091, 0.250      # width across fins, height, length (flow direction)
tb = 0.014                          # base plate thickness, each side (surface to fin root)
tf = 0.0016                         # fin (web) thickness
b = 0.00415                         # channel width
Nf, Nch = 36, 37                   # fins, channels
tw = (W - Nf*tf - Nch*b)/2         # side wall thickness (~6.9 mm)
Lf = Hs - 2*tb                    # fin length plate to plate (63 mm)
Ds = 0.100                         # SCR contact diameter
a = Ds/2; As = np.pi*a**2; Ap = W*L; rp = np.sqrt(Ap/np.pi)

# ---- materials ----
k = 200.0                          # Al 6060/6063 extrusion [W/(m K)]
rho_a, cp_a, k_a, nu_a, Pr = 1.127, 1007.0, 0.0271, 1.70e-5, 0.71 # air, 40 degC

Aff = Nch*b*Lf                    # free flow area
Afr = W*Hs
sigma = Aff/Afr
Dh = 2*b*Lf/(b+Lf)
Atop = Nch*b*L                    # channel top walls (at base temperature)

def h_channel(Vch):
    """Teertstra, Yovanovich, Culham (1999), h referred to inlet air temperature."""
    Reb = Vch*b/nu_a
    Res = Reb*b/L
    nu_fd = Res*Pr/2
    nu_dev = 0.664*np.sqrt(Res)*Pr**(1/3)*np.sqrt(1 + 3.65/np.sqrt(Res))
    Nu = (nu_fd**-3 + nu_dev**-3)**(-1/3)
    return Nu*k_a/b, Nu, Reb, Res

def fin_conductance(h, t, two_sided, tip_ratio):
    """Conductance [W/K] of one fin (thickness t, length Lf, depth L) whose tip is
    attached to the bottom plate; the plate is modelled as a convective tip with
    h_t = h*tip_ratio (tip_ratio = collector area / fin cross-section)."""
    P = 2*L if two_sided else L
    Ac = t*L
    m = np.sqrt(h*P/(k*Ac)); M = np.sqrt(h*P*k*Ac)
    z = h*tip_ratio/(m*k); mL = m*Lf
    G = M*(np.sinh(mL) + z*np.cosh(mL))/(np.cosh(mL) + z*np.sinh(mL))
    return G, mL

def G_total(h):
    Gf, mL = fin_conductance(h, tf, True, b/tf) # internal fins
    Gw, _ = fin_conductance(h, tw, False, (b/2)/tw) # side walls, inner face only
    G = Nf*Gf + 2*Gw + h*Atop
    eta_eff = Gf/(h*2*Lf*L)
    return G, mL, eta_eff

def lee_spreading(heff, kk=k):
    """Lee, Song, Au, Moran (1995): circular source (radius a) on a circular plate
    (radius rp, thickness tb) with uniform h_eff on the back. Returns R_max, R_avg, R_1D."""
```

```

eps, tau, Bi = a/rp, tb/rp, heff*rp/kk
lam = np.pi + 1/(eps*np.sqrt(np.pi))
Phi = (np.tanh(lam*tau) + lam/Bi)/(1 + lam/Bi*np.tanh(lam*tau))
psi_max = eps*tau/np.sqrt(np.pi) + (1-eps)*Phi/np.sqrt(np.pi)
psi_avg = eps*tau/np.sqrt(np.pi) + 0.5*(1-eps)*1.5*Phi
return psi_max/(kk*np.sqrt(As)), psi_avg/(kk*np.sqrt(As)), tb/(kk*Ap)

def pressure_drop(Vch):
    """Laminar developing flow in a rectangular duct (Shah & London) + contraction/expansion."""
    Re = Vch*Dh/nu_a
    al = b/Lf
    fRe = 24*(1 - 1.3553*al + 1.9467*al**2 - 1.7012*al**3 + 0.9564*al**4 - 0.2537*al**5) # Fanning
    xp = L/(Dh*Re)
    fapp = (3.44/np.sqrt(xp) + (fRe + 0.674/(4*xp) - 3.44/np.sqrt(xp))/(1 + 2.9e-5/xp**2))/Re
    Kc, Ke = 0.42*(1 - sigma**2), (1 - sigma**2)**2
    return 0.5*rho_a*Vch**2*(4*fapp*L/Dh + Kc + Ke)

if __name__ == "__main__":
    print(f"Lf={Lf*1e3:.0f} mm tw={tw*1e3:.1f} mm Aff={Aff*1e4:.1f} cm2 sigma={sigma:.3f} Dh={Dh*1e3:.2f} mm rp={rp*1e3:.1f} mm")
    print(" Q[m3/h] Vch[m/s] Re_b Nu_b h[W/m2K] eta_f G[W/K] R_conv R_sp,avg R_sp,max R_th,avg R_th,max dp[Pa]")
    for Vch in [1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 10]:
        h, Nu, Reb, Res = h_channel(Vch)
        G, mL, eta = G_total(h)
        Rmax, Ravg, Rld = lee_spreading(G/Ap)
        Q = Vch*Aff*3600
        print(f"Q:{Q*7.0f} {Vch:7.1f} {Reb:6.0f} {Nu:5.2f} {h:8.1f} {eta:6.3f} {G:7.2f} {1/G:8.4f} {Ravg:8.4f} {Rmax:8.4f} {1/G+Ravg:8.4f} {1/G+Rmax:8.4f} {pressure_drop(Vch):6.1f}")

```

B Finite-difference model script

Requires NumPy and SciPy; imports the file above. Run time about three minutes.

```

"""
Heatsink 778746 -- 3-D finite-difference thermal model (steady state and step response).
Top plate: 3-D (x, y, z); fins and side walls: 2-D (x, s); bottom plate: 3-D (x, y, z).
Uniform channel h from hs778746_analytic.h_channel, referred to inlet air temperature.
Run: python hs778746_fd.py (about 3 min; writes results_fd.json and foster.json)
"""

import json, time
import numpy as np
import scipy.sparse as sp
import scipy.sparse.linalg as spla
from scipy.optimize import curve_fit
from hs778746_analytic import (W, Hs, L, tb, tf, b, Nf, tw, Lf, a, Ap, As, Aff, k,
                               h_channel, G_total, lee_spreading, pressure_drop)

rho, cp = 2700.0, 900.0

def build(h, hj=np.inf, Nx=50, Nz=7, Ns=10, Nz=2, h_out=0.0):
    """Assemble conductance matrix K, capacitance vector C and unit-power vector P."""
    dx = L/Nx
    p = tf + b
    we = np + b/2 # edge cell: side wall + half channel
    wy = np.array([we] + [p]*Nf + [we]); Ny = len(wy)
    dz, ds, dzb = tb/Nz, Lf/Ns, tb/Nzb
    n = 0
    itp = np.arange(Nx*Ny*Nz).reshape(Nx, Ny, Nz); n += itp.size
    ifn = n + np.arange(Nx*Nf*Ns).reshape(Nx, Nf, Ns); n += ifn.size
    iwl = n + np.arange(Nx*2*Ns).reshape(Nx, 2, Ns); n += iwl.size
    ibp = n + np.arange(Nx*Ny*Nzb).reshape(Nx, Ny, Nzb); n += ibp.size
    N = n
    C = np.zeros(N); Gg = np.zeros(N); I = []; J = []; V = []
    link = lambda i, j, g: (I.append(i), J.append(j), V.append(g))
    # plates
    for idx, nz, dzz in ((itp, Nz, dz), (ibp, Nz, dzb)):
        for ix in range(Nx):
            for iy in range(Ny):
                for iz in range(nz):
                    nd = idx[ix, iy, iz]
                    C[nd] = rho*cp*dx*wy[ix]*dzz
                    if ix+1 < Nx: link(nd, idx[ix+1, iy, iz], k*wy[ix]*dzz/dx)

```

```

        if iy+1 < Ny:
            g = k*dx*dzz/(0.5*(wy[iy]+wy[iy+1]))
            if 1 <= iy <= Nf-1 and iy % 2 == 0 and np.isfinite(hj): # module joint every 2 fins
                g = 1/(1/g + 1/(hj*dx*dzz))
            link(nd, idx[ix, iy+1, iz], g)
        if iz+1 < nz: link(nd, idx[ix, iy, iz+1], k*dx*wy[iy]/dzz)

# fins
for ix in range(Nx):
    for jf in range(Nf):
        iy = jf + 1
        for s in range(Ns):
            nd = ifn[ix, jf, s]
            C[nd] = rho*cp*dx*tf*ds
            Gg[nd] += h*2*dx*ds
            if ix+1 < Nx: link(nd, ifn[ix+1, jf, s], k*tf*ds/dx)
            if s+1 < Ns: link(nd, ifn[ix, jf, s+1], k*dx*tf/ds)
            link(ifn[ix, jf, 0], itp[ix, iy, Nz-1], 1/((dz/2)/(k*dx*wy[iy]) + (ds/2)/(k*dx*tf)))
            link(ifn[ix, jf, Ns-1], ibp[ix, iy, 0], 1/((dzb/2)/(k*dx*wy[iy]) + (ds/2)/(k*dx*tf)))
            Gg[itp[ix, iy, Nz-1]] += h*dx*(wy[iy]-tf) # channel top wall
            Gg[ibp[ix, iy, 0]] += h*dx*(wy[iy]-tf) # channel bottom wall

# side walls (inner face convects)
for ix in range(Nx):
    for jw, iy in enumerate((0, Ny-1)):
        for s in range(Ns):
            nd = iwl[ix, jw, s]
            C[nd] = rho*cp*dx*tw*ds
            Gg[nd] += h*dx*ds + h_out*dx*ds
            if ix+1 < Nx: link(nd, iwl[ix+1, jw, s], k*tw*ds/dx)
            if s+1 < Ns: link(nd, iwl[ix, jw, s+1], k*dx*tw/ds)
            link(iwl[ix, jw, 0], itp[ix, iy, Nz-1], 1/((dz/2)/(k*dx*wy[iy]) + (ds/2)/(k*dx*tw)))
            link(iwl[ix, jw, Ns-1], ibp[ix, iy, 0], 1/((dzb/2)/(k*dx*wy[iy]) + (ds/2)/(k*dx*tw)))
            Gg[itp[ix, iy, Nz-1]] += h*dx*(wy[iy]-tw)
            Gg[ibp[ix, iy, 0]] += h*dx*(wy[iy]-tw)

if h_out > 0: # optional: outer face of bottom plate
    for ix in range(Nx):
        for iy in range(Ny):
            Gg[ibp[ix, iy, Nz-1]] += h_out*dx*wy[iy]

I, J, V = map(np.array, (I, J, V))
K = sp.coo_matrix((np.r_[V, V, -V, -V], (np.r_[I, J, I, J], np.r_[I, J, J, I])), shape=(N, N)).tocsr()
K = K + sp.diags(Gg)

# source: fraction of each top cell inside the circle of radius a
xc = (np.arange(Nx)+0.5)*dx - L/2
ye = np.r_[0, np.cumsum(wy)] - W/2
frac = np.zeros((Nx, Ny)); sub = 8
for ix in range(Nx):
    xs = xc[ix] - dx/2 + (np.arange(sub)+0.5)*dx/sub
    for iy in range(Ny):
        ys = ye[iy] + (np.arange(sub)+0.5)*wy[iy]/sub
        X, Y = np.meshgrid(xs, ys)
        frac[ix, iy] = np.mean(X**2 + Y**2 <= a**2)
Acell = dx*wy[None, :]*frac
P = np.zeros(N); P[itp[:, :, 0]] = (Acell/Acell.sum()).ravel()
q = P[itp[:, :, 0]]/(dx*wy[None, :]) # flux per unit power
geom = dict(itp=itp, ifn=ifn, iwl=iwl, ibp=ibp, Acell=Acell, q=q, dz=dz,
            xc=xc, yc=np.cumsum(wy)-wy/2-W/2, frac=frac, N=N, Gg=Gg)
return K, C, P, geom

def surface(T, geom):
    """Contact-face temperature per unit power: node value + half-cell correction."""
    return T[geom['itp'][:, :, 0]] + geom['q']*(geom['dz']/2)/k

def steady(K, P, geom):
    T = spla.spsolve(K.tocsc(), P)
    Ts = surface(T, geom); A = geom['Acell']
    return T, Ts, (Ts*A).sum()/A.sum(), Ts[A > 0].max()

def transient(K, C, P, geom, decades=(-3, 5), spd=90):
    """Implicit Euler step response; constant dt inside each decade (spd steps per decade)."""
    N = len(C); T = np.zeros(N); t = 0.0; Cd = sp.diags(C); A = geom['Acell']
    ts, Za, Zm = [], [], []
    for dec in range(*decades):
        dt = 10.0**dec/(spd/9)
        lu = spla.splu((Cd/dt + K).tocsc())
        for _ in range(spd):
            T = lu.solve(C*T/dt + P); t += dt

```

```

        Ts = surface(T, geom)
        ts.append(t); Za.append((Ts*A).sum()/A.sum()); Zm.append(Ts[A > 0].max())
    return np.array(ts), np.array(Za), np.array(Zm)

def foster_fit(t, Z, n=4, tmin=0.05):
    f = lambda t, *p: sum(np.exp(p[i])*(1-np.exp(-t/np.exp(p[n+i])))) for i in range(n))
    m = t >= tmin
    p0 = np.r_[np.log(np.full(n, Z[-1]/n)), np.log(np.logspace(-0.5, 2.5, n))]
    popt, _ = curve_fit(f, t[m], Z[m], p0=p0, sigma=Z[m]*0.02 + 2e-5, maxfev=40000)
    R, tau = np.exp(popt[n:]), np.exp(popt[n:]); o = np.argsort(tau)
    return R[o], tau[o], np.max(np.abs(f(t[m], *popt) - Z[m])/Z[m])

if __name__ == "__main__":
    out = {'sweep': [], 'joint': [], 'zth': {}}
    for Vch in [1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 10]:
        h, Nu, Reb, Res = h_channel(Vch)
        K, C, P, geom = build(h)
        T, Ts, Ravg, Rmax = steady(K, P, geom)
        G, mL, eta = G_total(h); Rsm, Rsa, _ = lee_spreading(G/Ap)
        out['sweep'].append(dict(V=Vch, Q=Vch*Aff*3600, Reb=Reb, Nu=Nu, h=h, eta=eta, G=G,
                                Ran_avg=1/G+Rsa, Ran_max=1/G+Rsm, Rsp_avg=Rsa, Rsp_max=Rsm,
                                Rfd_avg=Ravg, Rfd_max=Rmax, dp=pressure_drop(Vch)))
        print(f"V={Vch:4.1f} m/s  Q={Vch*Aff*3600:4.0f} m3/h  analytic {1/G+Rsa:.4f}/{1/G+Rsm:.4f}  FD {Ravg:.4f}/{Rmax:.4f} K/W")
    h = h_channel(4.0)[0]
    for hj in [np.inf, 5e4, 2e4, 1e4, 5e3]:
        K, C, P, geom = build(h, hj=hj); _, _, Ravg, Rmax = steady(K, P, geom)
        out['joint'].append(dict(hj=hj, Ravg=Ravg, Rmax=Rmax)); print("joint hj =", hj, Ravg, Rmax)
    K, C, P, geom = build(h, h_out=5.0); _, _, Ravg, Rmax = steady(K, P, geom)
    out['hout5'] = dict(Ravg=Ravg, Rmax=Rmax); print("h_out = 5:", Ravg, Rmax)
    K, C, P, geom = build(h); T, Ts, Ravg, Rmax = steady(K, P, geom)
    itp = geom['itp']; iy0 = np.argmin(np.abs(geom['yc'])); ix0 = np.argmin(np.abs(geom['xc']))
    out['prof_x'] = dict(x=(geom['xc']*1e3).tolist(), Tsurf=Ts[:, iy0].tolist(),
                        Troot=T[itp[:, iy0, -1]].tolist(), Tbot=T[geom['ibp'][:, iy0, 0]].tolist())
    out['prof_y'] = dict(y=(geom['yc']*1e3).tolist(), Tsurf=Ts[ix0, :].tolist(),
                        Troot=T[itp[ix0, :, -1]].tolist(), Tbot=T[geom['ibp'][ix0, :, 0]].tolist())
    Pc = geom['Gg']*T; mask = np.zeros(len(T), bool)
    for ix in range(len(geom['xc'])):
        for jf in range(Nf):
            if geom['frac'][ix, jf+1] > 0.5: mask[geom['ifn'][ix, jf, :]] = True
    out['share_under_source'] = float(Pc[mask].sum()); out['mass'] = float(C.sum()/cp)
    print("mass", out['mass'], "kg; share under footprint", out['share_under_source'])
    fits = {}
    for Vch in [2.0, 4.0, 8.0]:
        h = h_channel(Vch)[0]; K, C, P, geom = build(h); t0 = time.time()
        ts, Za, Zm = transient(K, C, P, geom)
        out['zth'][str(Vch)] = dict(t=ts.tolist(), Zavg=Za.tolist(), Zmax=Zm.tolist())
        for key, Z in (('Zavg', Za), ('Zmax', Zm)):
            R, tau, err = foster_fit(ts, Z)
            fits[f"{Vch}_{key}"] = dict(R=R.tolist(), tau=tau.tolist(), err=float(err))
            print(f"V={Vch} {key}: R_i [mK/W] = {np.round(R*1e3,2)}  tau_i [s] = {np.round(tau,2)}  max rel err {err*100:.1f}% ({time.time()-t0:.0f} s)")
    json.dump(out, open('results_fd.json', 'w')); json.dump(fits, open('foster.json', 'w'))

```